

Hyunggyu Min

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Research Interests

Statistical and computational genomics; single-cell, functional and regulatory genomics; high-throughput perturbation studies, MPRA and CROP-seq; statistical methodology and sequence data analysis.

Additional interests: dimension reduction, optimization and quantum computing.

Education

Oct 2023 - Present	University of North Carolina at Chapel Hill , Chapel Hill, NC PhD student in Biostatistics
Sep 2020 - Oct 2023	Yonsei University , Seoul, Republic of Korea M.A. in Statistics GPA: 4.27/4.30 (4.00/4.00)
Mar 2014 - Aug 2020	Yonsei University , Seoul, Republic of Korea B.A. in Applied Statistics Statistics & Mathematics GPA: 4.11/4.30 (3.91/4.00); Overall GPA: 3.70/4.30 (3.56/4.00)

Publications

Published

Massively parallel reporter assay investigates shared genetic variants of eight psychiatric disorders

Cell (2025). Co-author. [10.1016/j.cell.2024.12.022](https://doi.org/10.1016/j.cell.2024.12.022)

Altered gut microbiota in individuals with episodic and chronic migraine

Scientific Reports (2023). Co-author. [10.1038/s41598-023-27586-4](https://doi.org/10.1038/s41598-023-27586-4)

Effect of orally administered heat-treated *Lactobacillus plantarum* LM1004 on the innate immune system: A randomized, placebo-controlled, double-blind study

Journal of Functional Foods (2022). Co-author. [10.1016/j.jff.2022.105293](https://doi.org/10.1016/j.jff.2022.105293)

Preprint / Manuscript in Revision

Perturbation of genes linked to common schizophrenia risk variants identifies cilia programs

bioRxiv (2026). Second author. Manuscript in revision.
[10.64898/2026.06.09.731172](https://doi.org/10.64898/2026.06.09.731172)

Software & Algorithms

MSF - An algorithm for fast querying of similar sequences. In development; private repository.

ZiPert - Zero-inflated-model-based differential gene expression analysis for Perturb-seq data. In development; private repository.

MPRA barcode-element mapping - A long-read Nanopore pipeline for mapping MPRA barcodes to elements. [Public GitHub repository](#).

Research & Leadership Experience

Oct 2023 - Present	Won Lab, Department of Genetics, UNC-Chapel Hill <i>PhD Researcher</i> Research in single-cell and functional genomics, including in vivo MPRA and scMPRA. Development of statistical and computational methods for high-dimensional genomic data. Advisor: Prof. Hyejung Won wonlab.org
Jun 2022 - May 2023	Korea Quantum Computing, R&D Department <i>Researcher</i> IBM Hub activities for quantum computing outreach and research in quantum data science. Conducted quantum data science projects and delivered introductory lectures. Supervisor: Dr. Kion Kim
Mar 2022 - Feb 2023	Yonsei University, Quantum DnA Laboratory <i>Researcher</i> Research on quantum convolutional neural networks and quantum error mitigation. Advisor: Prof. Daniel Kyungdeock Park
Mar 2021 - Aug 2023	Yonsei University, Institute of Data Science <i>Lead Consulting Assistant</i> Led statistical consulting across biology, social science, engineering and other disciplines. Consulted on 20+ cases in 2021; led junior consultants and supervised projects in 2022. Director: Prof. SeungHo Kang
Jan 2019 - Dec 2020	Yonsei University, Data Science Lab (DSL) <i>Academic Coordinator</i> Founding member of an undergraduate academic society focused on data science and statistical theory. Designed study programs and coordinated projects. Advisor: Prof. TaeYoung Park

Teaching Experience

Yonsei University	Teaching Assistant Quantum Finance Computing (Sep - Dec 2022) Multivariate Analysis (Sep - Dec 2022; Sep - Dec 2021) Theory of Dimension Reduction (Mar - Jun 2022) Qiskit for Data Science (Mar - Jun 2022) Big Data and Quantum Computing (Sep - Dec 2021) Statistical Data Analysis (Mar - Jun 2021) Instructor: Prof. Hakbae Lee
Yonsei University	Teaching Assistant , Introduction to Statistics (Sep - Dec 2020) Instructor: Prof. Hogyu Lee
Korea Quantum Computing	Lecturer , Introduction to Quantum Data Science (Jan - Feb 2023) Director: Dr. Kion Kim

Programming Skills

Programming: Python (MuData, Dask, FAISS, scikit-learn, PyTorch/TensorFlow), R (Seurat, Rcpp, Shiny, tidyverse), SAS, Linux, Slurm.

Tools & platforms: Docker, Git, SQL, Notion, Slack.

Awards & Honors

Excellence Prize, 2019 Embrain Panel Big Data Analysis Competition (May 2019).
High Honors, Yonsei University (Fall 2019).